

Auteur: Rune Suy
Studierichting: Informatica
Oefeningenreeks 7E nr. 1.14

$$f(x, y) = x^3 + y^3 + 3xy + 3$$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y$$

$$\frac{\partial f}{\partial y} = 3y^2 + 3x$$

$$\begin{cases} 3x^2 + 3y = 0 \\ 3y^2 + 3x = 0 \end{cases}$$

$$\Leftrightarrow -x^2 = y$$

$$\Leftrightarrow 3x^4 + 3x = 0$$

$$\Leftrightarrow x(x-1)(3x^2 - 3x + 3)$$

$$x = 0 \Rightarrow y = 0$$

of

$$x = -1 \Rightarrow y = -1$$

$$H(x, y) = \det \begin{pmatrix} 6x & 3 \\ 3 & 6y \end{pmatrix} = 36xy - 9$$

$$H(0, 0) = -9 < 0 \text{ dus } (0, 0) \text{ is een zadelpunt}$$

$$H(-1, -1) = 27 > 0 \text{ en } \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} < 0 \text{ dus } (-1, -1) \text{ is een maximum}$$

References